

Mid-October 2025

The HUMAN Digest 1.13

Lake Forest College's AI newsletter, courtesy of the HUMAN grant



Image created in Midjourney by David Sanchez-Burr.

Hello.

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Image via NASA

The environmental impacts of AI are considerable, and the location of data centers can make the problem worse. U.S. based data centers directly used an estimated **17 billion gallons of water in 2023** according to an analysis undertaken by the Lawrence Berkley National Laboratory. The same study estimates that indirect water use via electricity production is an order of magnitude larger, at 211 billion gallons.

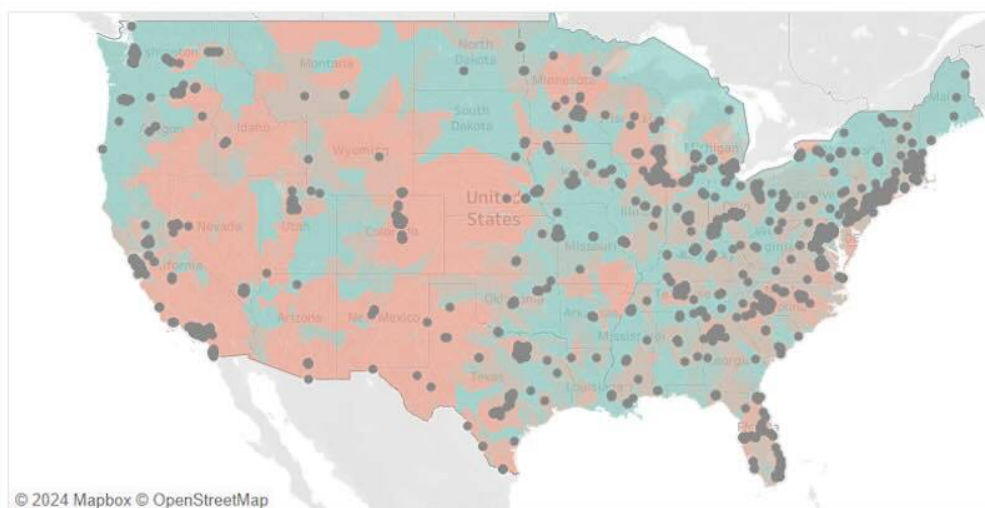
Banana for scale = Lake Michigan holds on the order of 1.7×10^{17} gallons of water (i.e. ~170 quadrillion gallons).

So, even with rapid AI/data-center growth, it would take several thousands years to consumptively use a “Lake Michigan” worth of water for cooling.

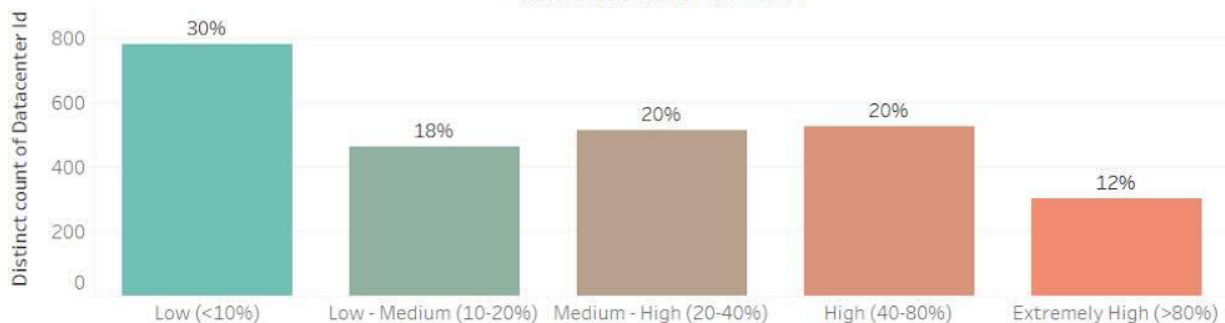
Similarly, the St. Lawrence Seaway’s outflow could supply the 2023 usage amount in a little over 14 hours.

Unfortunately, we are building some data centers in the U.S. in the [wrong places](#).

Location of >2500 Datacenters in United States



Wri Baseline Water Stress Label



The geography of data-center growth has followed cheap land and tax incentives more than water availability (although population density is a contributing factor). Many new facilities are being built in regions facing chronic drought such as the [desert southwest](#). By contrast, much of the upper Midwest, with abundant surface and groundwater remains under built.

The HUMAN Digest welcomes feedback at kee@lakeforest.edu

Justin Kee, editor-in-chief

Davis Schneiderman, Principal Investigator, [HUMAN](#) grant

AI in higher education

1. An article examining the AI arms race between [students and professors](#) in higher education. And it is [not going away](#).
2. What is the [truth](#) in the AI world? K-12 educators are struggling with issues regarding AI use similar to the issues seen in college classrooms.
3. [Reimagining](#) schools in the age of AI. The author's description of Adaptive Threshold Learning (ATL) which seeks to synthesize the role of the teacher as someone who facilitates active experience that leads to growth, along with the postmodern notion of idea the learning is a conversational partner in identity formation. These ideas are being explored by using AI to become more of a teacher than an oracle, as demonstrated in ChatGPT's [study mode](#).

General news

1. OpenAI released their video generation app [Sora 2](#). It can generate video, sound, speech and music in a (somewhat) coherent manner. Sora [contains](#) a slew of new features such as parental controls and a cameo function that allows those who opt-in to become characters in videos that others make. Sora 2 also functions like a video-based social media network and watermarks its content to indicate its AI origin. It is interesting to note OpenAI's stated [philosophy](#) towards the user's algorithmically curated slop i.e. feeds.
2. European AI [sovereignty](#) is a big question for the Continent. Chief Meta research scientist Yann LeCun has raised the issue of AI sovereignty, wherein the most powerful AI systems are not controlled by a [handful of companies](#) on the West Coast of the U.S. He noted that each nation should build their own AI systems that reflects their own culture and values. The running joke in S.F. is that "The U.S. innovates, China replicates and Europe regulates." (see the [E.U. AI Act](#).)
3. California's approach to [AI safety](#) via the Transparency in Frontier Artificial Intelligence Act. Transparency, safety frameworks and required disclosure for "frontier developers" (in excess of \$500 million in annual revenue) who are developing "frontier models" (defined as 10^{26} floating point or integer operations used in training).
4. Anthropic, the lab behind the popular Claude family of models, has given Claude the [agency to end a subset of conversations](#) when users persisted with harmful requests

(how do I commit a terroristic act) or repeatedly abused Claude.

Some perspectives

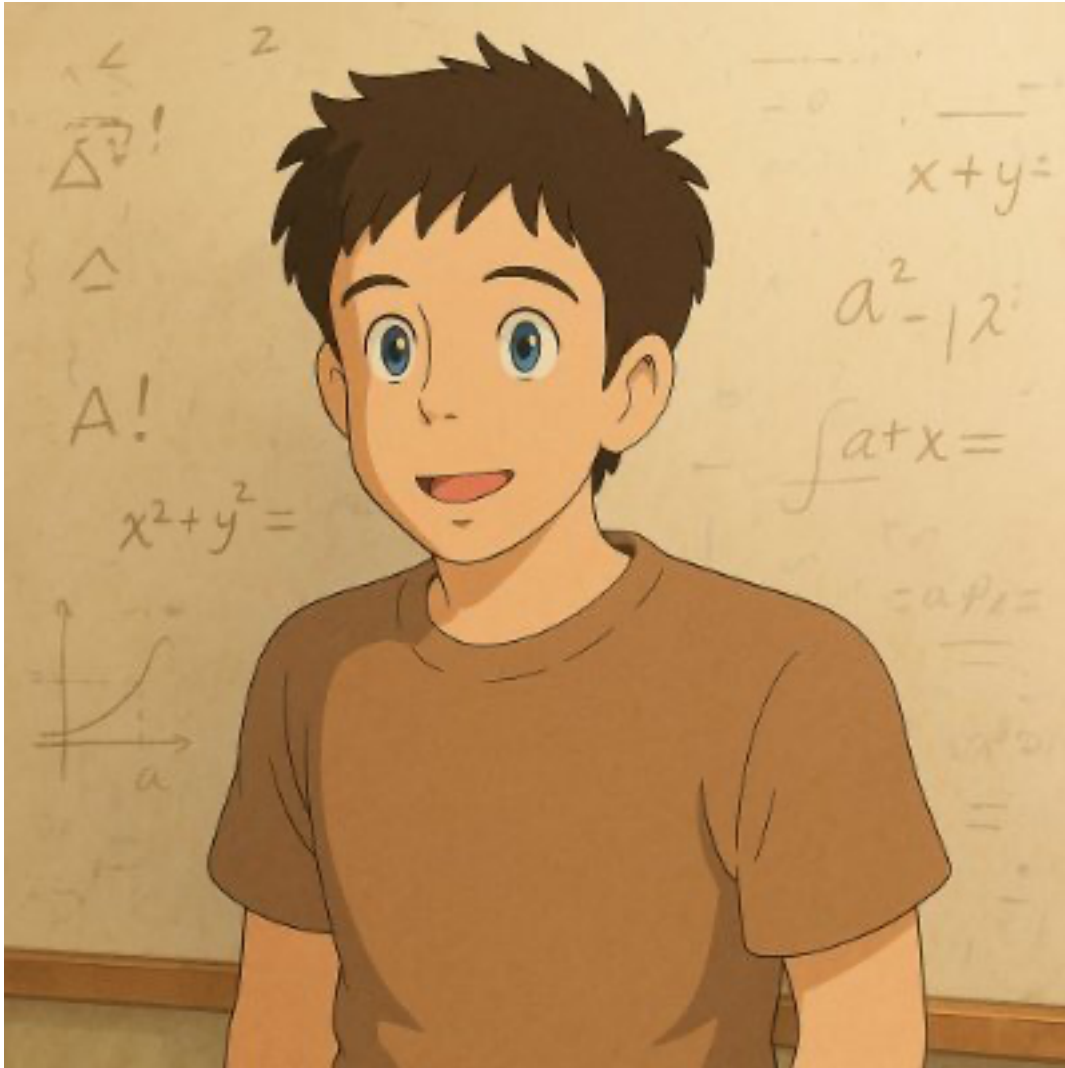
[Let's talk about AI art](#). A graphic novel perspective from Matthew Inman, the co-creator of [Exploding Kittens](#) as well as numerous books.

[Three perspectives](#) on consciousness.

A useful term

Model Collapse A phenomenon where AI systems that are [trained on data generated by other AIs](#) begin to degrade in quality over time, producing increasingly homogenized and inaccurate outputs. As synthetic content spreads through the internet this phenomenon raises questions about data provenance and long-term model sustainability. See also dead internet theory.

People to know



Sam Altman, born on April 22, 1985, in Chicago, Illinois, is one of the most influential figures in contemporary technology and artificial intelligence. Raised in St. Louis, Missouri, Altman's early interest in computers and programming led him to study computer science at Stanford University where he left without completing his degree. In his early twenties he co-founded Loopt, a pioneering location-sharing app launched in 2005. Despite initial enthusiasm and venture backing, Loopt was ultimately acquired for about \$43 million.

Image via X

Altman began working at [Y Combinator](#), a Silicon Valley venture capital firm. He initially joined as a part-time partner in 2011, he was appointed President in 2014, succeeding founder Paul Graham. In 2015, Altman co-founded OpenAI, alongside Elon Musk, Ilya Sutskever, Greg Brockman, and others, with a mission to ensure that artificial general intelligence (AGI) benefits all of humanity. Initially structured as a nonprofit research lab, OpenAI evolved under Altman's stewardship into a "capped-profit" company to attract capital while maintaining its ethical commitments. His relentless push for larger-scale compute and data infrastructure, positioned OpenAI at the center of what some have called the world's first "AI industrial revolution."

In late 2023, Altman [briefly lost his position](#) after a surprise ousting by OpenAI's board, only to be reinstated within days following widespread backlash from employees and investors.

Image via X.

Since then, OpenAI has reorganized their for-profit arm and currently has an estimated valuation of \$500 billion. And ChatGPT has an estimated [800 million](#) active weekly users.

Technical AI

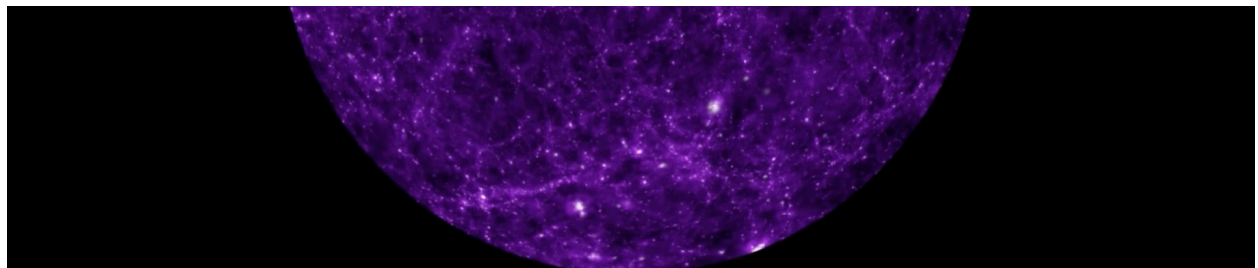
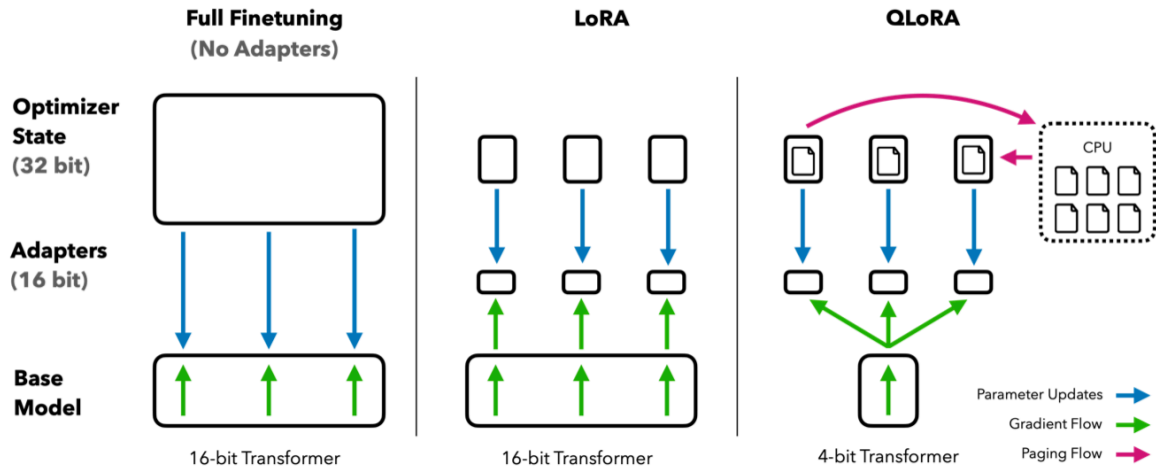
Efficient Fine-Tuning with QLoRA

As foundation models grow to hundreds of billions of parameters, retraining them from scratch becomes nearly impossible for most organizations. The technique of Quantized Low-Rank Adaptation (QLoRA) offers an efficient way to introduce new data. Introduced by Tim Dettmers and colleagues in 2023, [QLoRA](#) fine-tunes large language models using 4-bit quantization (reducing the number of



bits from 16 to 8 or as low as 2) while adding a small number of trainable low-rank adapter matrices. This approach preserves near full-precision performance at a fraction of the memory and energy cost. With QLoRA, a 65-billion-parameter model can be fine-tuned on a single 48 GB GPU.

Something to see



The cosmic [large scale structure](#) of the universe.

Image via Nasa

Just for fun

Mathematician [John H. Conway's](#) Game of Life, is a [simple, rule-based simulation](#) that can unexpectedly produce complex behavior. Martin Gardner popularized Conway's simulation in a 1970 edition of [Scientific American](#). It is a mathematical model that evolves over time into complex behavior according to a few deterministic rules.

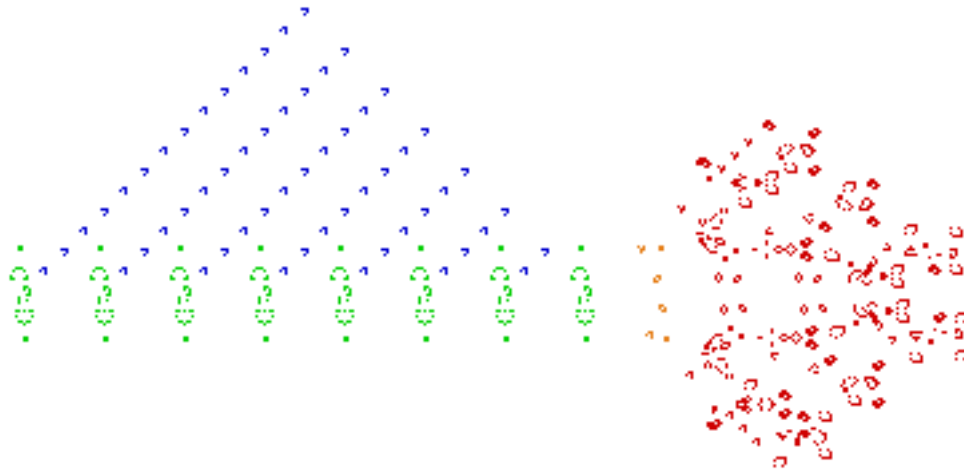


Image via Wikipedia

John H. Conway, one of three great [mathematicians](#) we lost in 2020.



HUMANS IN THE LOOP.

Have a tip or suggestion for an upcoming issue of the HUMAN Digest? Write Justin Kee an email at kee@lakeforest.edu. Thanks to ChatGPT and Claude for grammatical and thoughtful suggestions.

